

10

SOUND VARIATION IN BRAZILIAN PORTUGUESE

Valéria Neto de Oliveira Monaretto

This chapter presents an overview of the findings from descriptive studies on sound variation and change in Brazilian Portuguese (BP) in light of the Labovian sociolinguistic model. A brief historical retrospective of the first descriptions of BP is provided first, along with the emergence of databases for empirical research on spoken language. Next, we review some of the research findings on the consonant and vowel systems of BP, highlighting variables that are characteristic of specific Brazilian varieties according to differences in use in different regions of Brazil. Thus, this chapter provides the extent to which vowel and consonant variables are manifested in BP and identifies possible venues for future research in this area.

1 Introduction

Brazilian Portuguese (BP) has its own characteristics, which result from its historical evolution under Portuguese heritage. Different hypotheses have emerged with regard to its formation. These can be organized into three lines of research, namely, that which considers BP a continuation of Old Portuguese, with minor changes, that which views it as a new grammar developed from the 19th century onwards, and that which views it as emerging from a creolization process, resulting from a period as an African-based creole or semi-creole speech (cf. Castilho 2007: 13).

Regardless of the position one may take with regard to the origins of BP, it was important to know how this variety is actually manifested, through descriptions and empirical research. The first studies on BP came from Joaquim Mattoso Camara Jr., who, in 1934, penned his first articles (without signing them) in the paper *Correio da Manhã*. A number of his works on the Portuguese language emerged afterwards, culminating in his posthumous, yet unfinished, work *Estrutura da Língua Portuguesa* in 1970 (Uchôa 1975). In this volume, Camara Jr. describes aspects of Portuguese, such as phonemes, vowels, consonants, syllables, accentuation, and phonological vocabulary, all based on the variety spoken in Rio de Janeiro. Other early descriptive works of BP include those by Silva Neto (*A língua portuguesa no Brasil* 1960), Amadeu Amaral (*O dialeto caipira* 1920), Antenor Nascentes (*O falar carioca* 1922), and Nélson Rossi (*Atlas Prévio dos Falares Baianos* 1963), etc. Notwithstanding the essential contribution from these authors, Camara Jr. is unanimously considered the founding father of modern Brazilian linguistics (Altman 2012). Thus, most studies on sound change in BP followed the descriptive guidelines

from Camara Jr. Practice of dialectology in the 1950s and Labovian sociolinguistics afterwards provided much of the data source for subsequent studies and corpora. Specifically, they provided new directions to contemporary linguistics in Brazil based on rigorous data collection and well-defined methods.

Databases of spoken BP were first created in the 1970s under different motivations and objectives (Brandão 2022). The *Norma Urbana Culta* [Educated Urban Norm] (NURC) project emerged as part of an international project with the objective of analyzing the speech of individuals with higher education in five Brazilian metropolitan areas, namely, Rio de Janeiro, São Paulo, Recife, Salvador, and Porto Alegre, all of which had more than one million inhabitants at the time. A large database was created in 1972 based on formal utterances, informal dialogues between two speakers, and dialogues between documenter and speaker. These served as the basis for linguistics analysis in the *Projeto Gramática do Português Falado* [Spoken Portuguese Grammar Project] of the late 1980s.

With time and the rise and spread of Labovian sociolinguistics in the last 70 years, several corpora of spoken BP were created and several studies on the sound patterns of different BP speech communities have emerged. Thus, synchronic aspects of BP sounds were prioritized over diachronic aspects. Due to a lack of historical descriptions based on primary sources, in 1997 a nationwide project called *Para a História do Português Brasileiro* [Toward a History of Brazilian Portuguese] (cf. <https://sites.google.com/site/corporaphpb/home>) began to systematize data for the organization of a diachronic corpus of BP based on written language. Ten volumes were published from 2018 to 2021 as part of the collection *História do Português Brasileiro*. We highlight in this chapter much of the content of the third volume, which deals with sound change in BP (Hora et al. 2019).

Despite more than 50 years of research on BP, there are still regions and communities whose speech needs to be empirically investigated and described under quantitative sociolinguistic methods (Labov 1994; Weinreich et al. [1968] 2006). Speech sounds are one of the means through which many BP speakers perceive and characterize different varieties, as pioneered by Antenor Nascentes in the 1950s, who divided Brazil into five dialectal areas. As Callou (2002: 21) points out, this delimitation “collided with a great difficulty: the lack of determination of the so-called ‘isoglosses’, that is, demarcation lines for each of the linguistic phenomena that would singularize every dialect.” However, this does not prevent the acceptance of identifiable dialectal variants because

every Brazilian can intuitively recognize a great dividing line between ‘northern speech’ and ‘southern speech’: a ‘rhythm’ that is typical of northern speakers and another of southern speakers, pretonic open vowels from northeasterners and pretonic closed vowels from southerners, an assibilated *s* in the south as opposed to the *sh*-like sound from Rio de Janeiro, and the trilled *r* from Rio Grande do Sul as opposed to the *h*-like sound from Rio de Janeiro.

(Callou 2002: 21)

Some of these phonetic and phonological characteristics, perceived by Brazilian speakers, are innovative, while others are conservative in relation to European Portuguese (EP). Cunha (1986) indicates the following phenomena as the most widespread phonetic facts of BP (i.e., innovative variants that lack documentation in EP): the development of yod before coda-/s/ (e.g., *vez* [vejs] ‘time, turn’, *arroz* [a’hojs] ‘rice’); the vocalization of coda-/l/ (e.g., *anel* [a’new] and *alma* [’awma]); vowel epenthesis between medial consonant groups (e.g., *pneu* [pi’new] and *advogado* [adevo’gado]); and variation of open and closed pretonic vowels (e.g., *R[e]cife* /R[ɛ]cife ‘Recife’ and [o]linda/[ɔ]linda ‘Olinda’).¹

This chapter presents a state-of-the-art description of sound variation in current BP, focusing on consonants in §2, followed by vowels in §3. The former are described according to their position in relation to the vowel (i.e., whether prevocalic or postvocalic), while vowels are described in relation to stress (i.e., pretonic and posttonic). Concluding remarks and possible lines of inquiry for future research are provided in §4.

2 Consonants

Despite the descriptive problem posed by the distribution of consonants in BP, most variationist studies in Brazil have focused on consonant variation. This section presents, then, findings from research on consonantal variables from corpus data and the distribution of consonant variants in BP.

BP has an inventory of 19 consonant phonemes organized into obstruent and sonorant segments, which can be voiceless or voiced depending on their position within word. Consonant phonemes can be prevocalic and appear either at the beginning of the word (e.g., *roupa* ‘clothes’) or word-medially (e.g., *rouba* ‘he/she steals’). The exceptions are the so-called weak-r (i.e., the tap /ɾ/), the palatal lateral /ʎ/, and the palatal nasal /ɲ/ palatal, all of which appear only in intervocalic position in native words, such as in *aro* ‘hoop’, *alho* ‘garlic’, and *anho* ‘lamb’. The second segment in a prevocalic consonant cluster may only be /l/ or /ɾ/, e.g., *clave* ‘clef, key’, *crave* ‘fix -3SG.subj.’. In postvocalic position, liquids (e.g., *mar* ‘sea’, *mal* ‘evil’), non-labial fricatives (e.g., *paſta* ‘folder’, *rasgo* ‘I tear’, *folhas* ‘leaves’), and nasals are possible, the latter of which often nasalize the preceding vowel (e.g., *cantam* ‘they sing’, *bem* ‘well’). It is in this position where consonant variation often occurs in different BP varieties.

The most common process of variation involves r-sounds or rhotics, which have different realizations in internal (e.g., *carta* ‘letter’) and external (e.g., *mar* ‘sea’) coda position. There is an ongoing change in BP, which is already at advanced stage in some areas, where posterior variants (e.g., [h, x]) are replacing anterior ones (e.g., [r, ɾ]). The deletion of /ɾ/ in word-final position is also frequent in infinitive verb forms (e.g., *cantá* instead of *cantar* ‘to sing’). In onset position, there is variation in /r/-production. In bilingual areas of Italian descent in southern Brazil, strong-r is replaced by weak-r (e.g., [ɾ]ato instead of [r]ato ‘mouse’). In Portuguese-German bilingual areas, on the other hand, weak-r is replaced by strong-r (e.g., a[r]eia instead of a[ɾ]eia ‘sand’). Words like *rato* ‘mouse’ or *carro* ‘car’, for example, are both produced with a tap [ɾ] instead of a strong-r (trill or fricative). It is also possible to find a retroflex [ɻ] as the second element of a consonant group (e.g., *prato* ‘plate’) in inward areas of Midwestern states as well as in São Paulo, Paraná, and Minas Gerais, where this variant is often attested in coda position.

The variation in production of coda-/l/ distinguishes BP from EP by the predominant realization of the vocalized variant [w] in BP, a change process still in progress, albeit already consolidated in several areas. Other variants appear elsewhere, such as velarized [ɭ] or alveolar [l], mainly in Southern Brazil. Deletion may occur in some forms, such as in *azuØ* ‘blue’, *papeØ* ‘paper’, *jornaØ* ‘newspaper’, mainly in Northeastern BP. On the other hand, rhotacism (i.e., the replacement of /l/ by /r/ in coda position) is an older phenomenon process than /l/-vocalization. The replacement of /l/ by other variants, such as [j], [h], or Ø, is deemed ‘vulgar speech’ by Silva Neto (1970), although it is present in several parts of Brazil.

Coda-/s/ has three variants, namely, alveolar sibilants [s] and [z], palato-alveolar fricatives [ʃ] and [ʒ], and deletion Ø. Voiced or voiceless variants are performed according to the articulation of the consonant that follows, e.g., *ra[s]pa* ~ *ra[ʃ]pa* ‘shavings’, *ra[z]ga* ~ *ra[ʒ]ga* ‘he/she tears’. Alveolar sibilants are the most frequent and comprehensive variants, while palato-alveolars occur in

specific coastal varieties, such as in Rio de Janeiro, Recife, Salvador, and Florianópolis. A phenomenon that is related to coda-/s/ in particular is the occurrence of a variable glide [j], mainly in stressed syllables at the end of a word, as in *arro[j]z* ‘rice’ and *mê[j]s* ‘month’. According to Révah (1958), it is a particular feature of BP and occurs in select varieties. An inverse phenomenon (i.e., [j]-deletion) monophthongizes falling diphthongs followed by a coda fricative, such as *mas* for *mais* ‘more’ and *depôs* for *depois* ‘after’, which is observed both in EP and in Florianópolis, in Southern Brazil.

With regard to postvocalic nasals, variable vowel denasalization and monophthongization occurs in final unstressed syllables in BP, e.g., *homem* ~ *homi* ‘man’, *garagem* ~ *garage*, ‘garage’, *cantaram* ~ *cantaru* ‘they sang’.

Dental stops /t/ and /d/ are variably palatalized when followed by [i, ɪ, ʔ, j], being produced as [tʃ] and [dʒ], respectively, in most metropolitan areas in Brazil, as in *tia* ‘aunt’, *dia* ‘day’, *bote* ‘boat’, *pode* ‘can’, *tinta* ‘paint’, *pátio* ‘patio’. In several varieties of Northeastern BP, however, /t/ and /d/ are palatalized by a preceding glide [j], as in *oito* [oʃo] and *jeito* [ʒetʃo] ‘way’. In informal BP, the gerund suffix {-ndo} gets variably reduced to {-no} with the deletion of /d/, e.g., *cantando* ~ *cantano* ‘singing’, *esperando* ~ *esperano* ‘waiting’.

The following subsections present the findings from sociolinguistic studies focusing on variable /R/, /l/, and /S/ in coda position in BP.

2.1 Variable coda-/R/

/R/ is one of the consonants that presents a wide range of phonetic realizations. Indeed, it is one of the features that differentiates regional BP varieties. Most speakers realize it with a velar [x], uvular [χ], or glottal [h], although multiple other variants occur, such as tap [r], retroflex [ɽ], approximants [ɹ, j], and deletion. Word-final coda is the position that displays the greatest number of variants, all represented by the grapheme <r>.² The occurrence of these variant forms in coda position is linked to both social and linguistic factors.

A variationist analysis of /R/ in word-medial and word-final coda, based on data from the NURC project, suggests the following variant distribution. In Porto Alegre and São Paulo, the tap [r] predominates in the medial position, while the deletion (Ø) predominates in final position. In Rio de Janeiro and Salvador, a fricative variant ([x] or [h]) has the highest percentage word-internally, whereas deletion does so word-finally, being stable in Rio de Janeiro and in progress in Salvador. In Recife, the glottal fricative [h] is preferred.

In Southern BP (i.e., comprising the states of Rio Grande do Sul, Santa Catarina, and Paraná), Monaretto (1997, 2010) analyzes data from the *Variação Linguística do Sul do País* [Linguistic Variation in Southern Brazil] (VARSUL) project. Results suggest that, in terms of frequency, the tap predominates in word-medial coda in seven of the ten cities analyzed in the region. The velar fricative [x] predominates in Florianópolis, which is a city of Azorean colonization in the state of Santa Catarina. The trill [r] predominates in Flores da Cunha, a city of Italian colonization in Rio Grande do Sul, while a retroflex approximant [ɽ] is most frequent in Londrina, in northern Paraná. In relation to other places, there is a high percentage of retroflex variants in the states of São Paulo and Minas Gerais. This realization is observed mainly in inland São Paulo (cf. Guiotti 2002; Leite 2010; Oushiro & Mendes 2013; Rennicke 2011).

In Northern and Northeastern Brazil, the glottal fricative variant [h] and phonetic zero Ø present the highest percentage of occurrences, with variants [j] and [w] occurring in the state of Paraíba, and the palato-alveolar [ʃ] occurring in Piauí (cf. Hora & Monaretto 2003; Skeete 1996).

The examples in (1), adapted from Hora (2010: 62), summarize the variants of coda-/R/ in BP for the words *porta* ‘door’, *mar* ‘sea’, *garfo* ‘fork’, and *artista* ‘artist’.

- (1) [r] *porta, mar*
 [x] *porta*
 [t] *porta*
 [h] *porta*
 [j] *porta*
 [w] *garfo*
 [ʃ] *artista*
 Ø *garfo, mar*

Among the different realizations of /R/, deletion has received more attention from researchers because it reflects a potential last stage of /R/-weakening ([i.e., [r] → [x] → [h] → Ø), as well as the simplification of a CVC syllabic structure to a CV structure (cf. Callou 1987, 2008; Callou & Serra 2012; Callou et al. 1996; Hora & Monaretto 2003; Monaretto 2002; Oliveira 1983; Schwindt & Chaves 2019; Votre 1978; among others).

The deletion of /R/ at the end of a word is an old phenomenon in Portuguese and occurs more frequently in infinitive verb forms than in non-verb forms. It is common in spoken BP and does not seem to be stigmatized or conditioned by social factors. The difference among dialectal areas is with regard to its frequency in non-verb forms, as well as the progress of /R/-weakening and its realization as a fricative form, culminating afterwards in deletion. Prosodic structure with regard to the intonational phrase possibly disfavors /R/-weakening (cf. Callou & Serra 2014).

In Rio de Janeiro and Salvador, where fricative variants predominate, coda-/R/ deletion is also at an advanced stage, particularly in verb forms. While in Rio de Janeiro coda-/R/ deletion does not occur in non-verb forms, in Salvador it is almost categorical (cf. Callou et al. 1996). In the speech of João Pessoa in Northeastern Brazil, data from the *Variação Linguística da Paraíba* [Linguistic Variation in Paraíba] (VALPB) project point toward variable cases of deletion and glottal forms in word-medial position when followed by fricatives, such as in *ga[h]fo* ~ *ga(Ø)fo* ‘fork’, *ce[h]veja* ~ *ce(Ø)veja* ‘beer’ (Hora & Monaretto 2003: 128). In this position, there seems to be a weakening process that develops toward deletion. Indeed, /R/-weakening and deletion is notorious in Northeastern BP and takes place in verb and noun forms. Data from the *Atlas Linguístico do Brasil* [Linguistic Atlas of Brazil] (ALiB) corroborate these findings, as illustrated in Table 10.1.

The lower percentages of coda-/R/ deletion in Teresina and Aracaju is accounted for by the light contact that took place between Portuguese and African languages in these cities as compared to other Northeastern cities.

In Southern Brazil, deletion occurs mainly in word-final position in infinitive verb forms. Data from the VARSUL project points toward a regularity in the findings. Leal (2016) summarizes the results from previous studies following the data comparison model by Bailey and Tillery (2004) and offers an overview of /R/-deletion in word-final position, as shown Table 10.2.

Table 10.1 Coda-/R/ deletion in Northeastern capital cities (Callou et al. 2015: 207).

Percentage of deletion in verb forms	Percentage of deletion in non-verb forms	Northeastern Capital City
97–98%	83–95%	João Pessoa, Salvador, Maceió, and Fortaleza
94–98%	62–71%	Natal, Recife, and São Luís
83–85%	53–58%	Teresina and Aracaju

Table 10.2 Research on /R/-deletion in word-final position in verb and non-verb forms, using Southern BP data (adapted from Leal 2016: 308).

Author	Verb forms	Non-verb forms	City	Year of data collection
Monaretto (2002)	36%	1%	Porto Alegre	1970
Monaretto (2002)	80%	6%	Porto Alegre	1989
Bertani (1998)	97%	----	Porto Alegre	1988–1996
Monaretto (2000)	80%	10%	Curitiba Florianópolis	1988–1996
			Porto Alegre	
Gregis (2001)	96%	8%	Porto Alegre	1988–1996
Silveira (2010)	68%	3%	Blumenau Lages	1988–1996
			Londrina	
			Pato Branco	
Monaretto (2002)	76%	2%		1999

As illustrated in Table 10.2, the percentages of /R/-deletion in word-final position differ greatly between verb forms (e.g., *cantar* ‘to sing’, *vender* ‘to sell’) and non-verb forms (e.g., *açúcar* ‘sugar’, *revolver* ‘revolver’). The lower percentage of deletion (i.e., 36%) found for verb forms in Monaretto (2002) probably stems from the fact that the data for that study came from the NURC project from 1970, thus representing the speech of informants with a higher level of education, unlike the data and informants from other studies. In cities located in the interior of Southern states, Silveira (2010) reports that coda-/R/ is preserved at a higher percentage (i.e., 68%) than in the capital cities surveyed.

Despite deletion of coda-/R/ occurring more frequently in verb forms, Oliveira (1983) argues that the preservation of stress on the final syllable of the word guarantees that speakers identify those forms as infinitives (e.g., *cantá* for *cantar* ‘to sing’). In non-verb forms such as nouns and prepositions (e.g., *amor* ‘love’, *dor* ‘pain’, *por* ‘by, for’), coda-/R/ deletion seems to be restricted to a few words due to the rarity of the process in Southern BP varieties. Perhaps these cases could be more accurately examined under the precepts of exemplar theory to test for lexical frequency effects. Schwindt and Chaves (2019: 143) posit that the variable deletion rule of coda-/R/ versus the occurrence of other variants, and the debuccalization process of /r/ (which yields fricative variants and phonetic zero), represent divergent outputs that result in convergent outputs, that is, both are “two processes or two ordered grammars that compete distinctly at a certain surface level with the deletion of the final rhotic.”

In sum, the consonant /R/ presents a range of variants that are conditioned mainly by its position within the syllable and the dialectal area. The role played by other (mainly social) conditioning factors is not yet so clear. However, there is undoubtedly a predominance of posterior variants in most BP varieties, followed by deletion in coda position.

2.2 Variable coda-/l/

Coda-/l/, as in *sal* ‘salt’ and *salgado* ‘salty’, has different realizations in BP, ranging from dental or alveolar [l] and velarized [ɫ], both of which are restricted to specific dialects, to a vocalized variant [w], which is widespread in BP. The posteriorization of the lateral articulation is a fact of the historical evolution of Portuguese. Indeed, historically, it likely represents an intermediate stage before the deletion of intervocalic /l/ in pre-literary Portuguese, e.g., Latin *SALŪTEM* > *saúde* ‘health’ (cf. Williams 1975). Camara Jr. (1971a: 17) points out the change in the place of articula-

tion from dental to velar as a typical positional variant of the lateral consonant, also observed in the speech of Piracicaba (São Paulo) and in inland Paraná (cf. Secco 1977). In Belo Horizonte, the velar variant disappeared and was replaced by a vocalized variant [w] (Oliveira 1983).

The alveolar pronunciation [l] is not anomalous in coda position in the extreme south of Brazil (Camara Jr. 1977), particularly areas of Italian and German heritage, although there seems to be a shift among young people in favor of [w] (Battisti & Moras 2016). Indeed, in Southern BP, alveolar and velarized realizations are frequent and, in some places, predominant (cf. Espiga 2001; Quednau 1993; Tasca 1999). Tasca (2002) sees these realizations under telescoping, which can be observed currently in the varieties spoken in Rio Grande do Sul, where alveolar [l], velarized [ɫ], velarized-labialized [ɫw], and vocalized [w] all occur as variants of coda-/l/.

With regard to BP as a macro variety, however, the most frequent variant of coda-/l/ is the glide [w], which “cannot be considered a feature that is restricted to some areas, such as Ceará and Rio de Janeiro, as one would think judging by classical descriptions. When one compares the educated speech from São Paulo, Rio de Janeiro, Salvador, Recife, and Porto Alegre, it is easy to notice that Porto Alegre distinguishes itself from the other cities, with the alveolar/velar variant being realized almost exclusively in this city” (Leite & Callou 2002: 47).³

A comprehensive view of the status of the alveolar and velarized variants in Southern BP region is provided by Collischonn and Quednau (2010), who combined and reanalyzed data from Quednau (1993) for purposes of comparison and expansion of findings, as shown in Table 10.3.

Despite the lack of uniformity with regard to the number of informants, we notice the presence of the velarized variant in all the Southern cities surveyed, while the alveolar variant is found in four of the six cities, and vocalized variant predominates in the cities of the state of Paraná. In São José do Norte (in Southern Rio Grande do Sul), the velarized variant competes with the vocalized one. Taps and deletions are also found to be variants, albeit with lower frequencies of occurrence. Collischonn and Quednau (2010: 135) argue that there are two types of competition among coda-/l/ variants: “one between lateral and rhotic variants, and another between lateral and vocalized variants. For example, in São José do Norte, out of the 8 informants who vocalize the most, no rhotic occurrence was found in the speech of 6 of them. On the other hand, for those who do vocalize, we find lower⁴ rates of vocalization.” With regard to the linguistic factors that promote coda-/l/ vocalization, word-final position was found to favor this process more than word-medial position does, both in monosyllabic words (e.g., *sal* ‘salt’) and in posttonic final syllables (e.g., *fácil* ‘easy’). With regard to extralinguistic factor, location and young speakers favor the vocalized variant.

Table 10.3 Distribution of coda-/l/ variants in Southern BP, based on data from the VARSUL project. (Collischonn & Quednau 2010: 134; PR = Paraná; RS = Rio Grande do Sul)

Place	Number of informants	Number of occurrences	Vocalization	Tap	Deletion	Alveolar lateral	Velarized lateral
Londrina (PR)	8	997	80%	--	--	--	20%
Pato Branco (PR)	8	625	91%	2%	1%	--	6%
Curitiba (PR)	8	543	81%	1%	8%	4%	6%
Irati (PR)	16	1161	63%	3%	3%	8%	23%
Lages (PR)	16	798	51%	6%	4%	20%	16%
São José do Norte (RS)	16	1048	43%	1%	9%	15%	30%

Table 10.4 Percentages of word-internal (3,000 tokens) and word-final (6,000 tokens) coda-/S/ variants according to data from the NURC project (adapted from Callou 2015: 42).

<i>Coda-/S/ variants</i>								
	<i>Palato-alveolar</i>		<i>Glottal</i>		<i>Alveolar</i>		<i>Deletion</i>	
	<i>internal</i>	<i>final</i>	<i>internal</i>	<i>final</i>	<i>internal</i>	<i>final</i>	<i>internal</i>	<i>final</i>
Rio de Janeiro	90%	75%	6%	10%	1%	8%	2%	8%
São Paulo	9%	5%	0%	0%	88%	91%	3%	3%
Porto Alegre	23%	2%	0%	0%	77%	96%	0%	1%
Recife	84%	54%	5%	7%	10%	34%	2%	5%
Salvador	56%	31%	4%	9%	39%	51%	1%	9%

2.3 Variable coda-/S/

The consonant /S/ is variable in coda position only in select areas and cities in Brazil. The most general variant appears to be alveolar [s, z], followed by palato-alveolar [ʃ, ʒ], glottal [h], and phonetic zero (Ø). A contrast between Southern and Northern varieties is observed, with the former displaying alveolar variants and the latter showing palato-alveolar variants, which occur more frequently in word-internal codas (Leite & Callou 2002: 46). Table 10.4 summarizes data from the NURC project on the distribution of coda-/S/ variants in educated speech from five Brazilian cities.

As shown in Table 10.4, palato-alveolar and alveolar realizations occur in all cities. Palato-alveolars are more frequent in word-internal coda (e.g., *paʃta* ‘folder’) in the cities of Rio de Janeiro, Recife, and Salvador. In the cities of Porto Alegre and São Paulo, alveolar variants are preferred, particularly in word-final position (e.g., *arroz* ‘rice’).

Real-time studies show that palatalization process of coda-/S/ in the city of Rio de Janeiro is at an advanced stage. In the interior of the state of Rio de Janeiro, however, Rodrigues (2001) finds that the alveolar variant is predominant among men with low education, despite the increasing frequency of the palato-alveolar variant in communities that are more in contact with the capital city.

In sum, we have, consonants /R/, /l/, and /S/ vary considerably in BP varieties and are currently undergoing a process of posteriorization in coda position, particularly /l/. Southern BP varieties retain older variants, such as alveolar [r], [l] and [s], despite the advanced stage of vocalization of the lateral consonant.

3 Vowels

The BP vowel system has seven phonemes in its inventory in stressed position, namely, /a, ε, e, i, ɔ, o, u/. In unstressed positions, the number of vowels is reduced by neutralization. In pretonic position, there are five vowels, i.e., /a, e, i, o, u/. In posttonic non-final position, there are four vowels, i.e., /a, e, i, u/, and in final posttonic position, there are three vowels, i.e., /a, i, u/ (Camara Jr. 1975).

For Camara Jr. (1975), there are no phonemic nasal vowels in Portuguese. Vowel nasalization results from an underlying nasal consonant that can be in the same syllable (i.e., *mundo* ‘world’) or in the next one (e.g., *jannela* ‘window’). The former is categorical (cf. *mundo* ‘world’ vs. *mudo*

‘mute’), while the latter is not, so nasalization may or may not affect the preceding vowel, such as in *cama*, *janela*, *banana*, etc.

Word-final nasal diphthongs can be either stressed (e.g., *limão* ‘lime’) or unstressed (e.g., *limam* ‘they file’). The latter undergo a process of variable reduction and denasalization (e.g., *cantaram* ~ *cantar*[u] ‘they sang’, *homem* ~ *hom*[i] ‘man’), particularly by speakers with less education (Battisti 2002; Chaves 2017; Guy 1981; Votre 1978). Education may also be responsible for blocking the spread of this process, given the influence from the spelling of *-ão*, *-am*, and nouns ending in *-gem* (e.g., *garage* ‘garage’, *viagem* ‘trip’, etc.) (Schwindt 2010: 29).

Oral diphthongs can be either rising (e.g., *suar* ‘to sweat’, *piidade* ‘pity’) or falling (e.g., *peixe* ‘fish’, *pauta* ‘docket’). Rising diphthongs vary with hiatus (e.g., *suar* ~ *su.ar* ‘to sweat’), except for the sequences [kw] (e.g., *quatro* ‘four’) and [gw] (e.g., *água* ‘water’), both of which are derived from Latin. Some falling diphthongs, on the other hand, undergo a process of variable monophthongization (e.g., *peixe* ~ *pexe* ‘fish’), with no apparent sociolinguistic motivation.

Some scholars posit the existence of phonological and phonetic diphthongs. The former would be ‘true’ diphthongs and have two vowels (VV) in their underlying structure, e.g., *reitor* ‘dean’, *pauta* ‘docket’. Phonetic diphthongs, on the other hand, would be ‘false’ diphthongs and have only one underlying vowel, with a glide at the surface level, e.g., *peixe* ‘fish’, *feira* ‘market, fair’. These would be represented by falling diphthongs, which would be linked to a single element V in a light syllable CV, hence, their frequent monophthongization. This is the case of diphthongs /ei/, /ai/, /ou/ in front of a tap [ɾ] and a palatal or palato-alveolar consonant (e.g., *beira* ‘edge’, *caixa* ‘box’, *couro* ‘leather’), in which a glide appears on the surface by assimilation to the vowel feature of the following consonant (Bisol 1989, 1994). Data from spoken BP corroborates Bisol’s (1994) phonological analysis for some varieties, in which high percentage rates of monophthongization are found before [ɾ] and palatal consonants. The diphthong *ai* is the one least affected by monophthongization, followed by *ei* and *ou*, the latter of which seems to be the one most affected. The diphthong *ei*, on the other hand, is the one with a different distribution in BP varieties and its monophthongization has been widely studied (Araújo 1999; Cabreira 1996; Cristofolini 2011; Paiva 1996; Toledo 2011; among others).

Diphthongization also occurs in BP by means of glide epenthesis in final stressed syllable followed by /S/ (e.g., *três* → *tre[j]s* ‘three’, *arroz* → *arro[j]s* ‘rice’) or a palato-alveolar fricative. In the latter case, a glide is inserted despite the lack of a corresponding grapheme (e.g., *faxina* → *fa[j]xina* ‘cleaning’, *bandeja* → *bande[j]ja* ‘tray’). Diphthongization also emerges from the vocalization of coda-/l/, e.g., *mal* [maw] ‘evil’, *Brasil* [bra'ziw] ‘Brazil’, etc. (Révah 1958).

Unstressed closed mid vowels /e/ and /o/ also exhibit much variation in BP. Depending on their position within the word and the variety in question, they may be realized as [e] and [o], [ɛ] and [ɔ], or [i] and [u], respectively. Some linguistic factors favor the use of some of these variants, as is the case of high vowels in stressed syllables, which trigger a process of assimilation into pretonic vowels called ‘vowel harmony’ (e.g., *m[e]nino* ~ *m[i]nino* ‘boy’). Underlyingly, mid vowel opening features are delinked and connected to the opening features of the following vowel (Bisol 1981). Mid vowel raising may also occur when initial /e/ precedes a sibilant or nasal consonant (e.g., [i]*spada* ‘sword’, [i]*stranho* ‘strange’, [i]*mpregada* ‘maid’, [i]*nxada* ‘hoe’) and when there is no apparent phonetic or phonological motivation (e.g., [e]*levador* ~ [i]*levator* ‘elevator’, g[o]*verno* ~ g[u]*verno* ‘government’).

Other vowel processes affect syllable structure, such as vowel syncope in proparoxytones (e.g., *abobra* ~ *abóbora* ‘butternut squash, pumpkin’, *arvri* ~ *árvore* ‘tree’), vowel epenthesis in consonant groups (e.g., *pineu* ~ *pneu* ‘tire’, *adevogado* ~ *advogado* ‘attorney’), external sandhi triggering

vowel diphthongization or deletion (*menina humilde* ~ *meni[naw]milde* ~ *meni[nu]milde* ‘humble girl’) (cf. Amaral 2000; Bisol & Collischonn 2002; Cagliari 2007; Lee 2012; among others).

Variable mid vowels are discussed in further detail in §3.1, given the historical change they have undergone and the fact that they represent one of the features that most distinguishes BP from EP.

3.1 Final unstressed mid vowels

Unstressed mid vowels /o/ and /e/ undergo raising in word-final position in most BP varieties, with the predominance of variants [u] and [i], respectively (e.g., *bolo* ‘cake’, *quente* ‘hot’). The origins of this raising process remain controversial and one of the most difficult problems in the history of Portuguese (Monaretto 2013, 2019; Naro 1973).

Despite little textual evidence about the use of these variants, final unstressed mid vowel raising is widespread in BP, with the exception of a few Southern varieties (Milesky 2013). Table 10.5 provides data from Southern capital cities, where raising is high, with the exception of Curitiba, which is rather conservative and does not favor this process.

Among the linguistic factors that condition the variation of final unstressed mid vowels, research findings suggest that the preceding context plays a significant role. Specifically, segments that are [+high] tend to favor the raising of /e/ to [i], but they disfavor the raising of /o/ to [u]. Additionally, syllables closed by a sonorant coda also inhibit raising (e.g., *imóvel* ‘property’, *caráter* ‘character’).

3.2 Non-final unstressed mid vowels

Non-final mid vowels undergo two types of variable processes in BP, namely, (i) raising without a high vowel in the preceding syllable (e.g., *pêssego* [ˈpesigo] ‘peach’, *brócolis* [ˈbrɔkulis] ‘broccoli’) as well as by assimilation to a high vowel in the preceding syllable (e.g., *número* [ˈnumirɔ] ‘number’, *âncora* [ˈɛkura] ‘anchor’); and (ii) deletion, which triggers other segmental processes and produces changes to syllable structure (e.g., *pêssego* [ˈpeskɔ] ‘peach’, *véspera* [ˈvɛsprɐ] ‘eve’). The latter case refers to the reduction in the number of syllables from three to two, a phenomenon that has been productive since Latin and is still currently ongoing in BP (cf. Amaral 2002; De Paula 2010; and others). Non-final mid vowel raising patterns in BP also presents a situation that goes against Camara Jr.’s (1971b) initial proposal of four vowels in this position (i.e., /a, e, i, u/). Unlike the Rio de Janeiro variety, however, Southern BP provides evidence for the presence of [o] in this position as well, in words such as *fôsf[o]ro* ‘match’, *antíd[o]to* ‘antidote’, among others.

Table 10.5 Raising of /e/ and /o/ in final posttonic position in the capital cities of Southern Brazil (Adapted from Vieira 2010: 52, 56).

	City	Application/Total	%	Relative Weight
/e/-raising	Curitiba	1056/1970	53	0.14
	Florianópolis	1434/1568	91	0.61
	Porto Alegre	2272/2424	93	0.77
Total		4762/5962	79	
/o/-raising	Curitiba	2140/2750	81	0.22
	Florianópolis	2180/2206	95	0.55
	Porto Alegre	2730/2806	97	0.74
Total		7050/7622	90	

Table 10.6 Raising of /e/ in non-final posttonic position in the three Southern Brazilian states (adapted from Vieira 2010).

	State	Application/Total	%	Relative Value
/e/-raising	Rio Grande do Sul	131/165	79	0.63
	Santa Catarina	157/202	78	0.49
	Paraná	116/168	69	0.39
Total		404/535	76	

Vieira (2010) analyzes non-final unstressed (posttonic) vowels in cities of Southern Brazil. In the case of /o/, its frequent deletion prevents any possible analysis. However, results obtained for non-final posttonic /e/ suggest that raising of this vowel is not favored in cities of Paraná, as illustrated in Table 10.6.

Vieira (2010) also observes that the presence of a high vowel in the word favors the raising of non-final mid vowels /e/ and /o/ by regressive assimilation (e.g., *agrícola* ‘agricultural’).

In the city of Rio de Janeiro, De Paula (2010: 86) reports that, in popular speech, /e/-raising is almost categorical, while in educated speech raising is predominant, with 90% frequency. The author also reports several cases of deletion of non-final unstressed /o/.⁵

In Belo Horizonte, Ribeiro (2007) interprets the results of a variationist analysis on the behavior of non-final posttonic mid vowels under the precepts of Lexical Diffusion, through which changes are understood to be lexically abrupt and phonetically gradual.⁶ The lexicon takes on a role and assigns ‘life’ to the word. Certain words can present a certain behavior in processes of variation, as is the case with the near categorical raising of /o/ in *abóbora* and the non-raising of /e/ in *véspera* ‘eve’. Unlike Southern BP varieties and that of Rio de Janeiro, the raising of /o/ occurs more frequently than that of /e/ in Belo Horizonte.

3.3 Pretonic vowels

One of the oldest phonological phenomena in the development of Portuguese, already attested in spoken Latin, is the harmonization of pretonic mid vowels, which consists of replacing /e, o/ with the high vowels [i, u], respectively, when followed by another high vowel (e.g., *menino* ~ *minion* ‘boy’, *coruja* ~ *curuja* ‘owl’). Vowel harmony is a part of BP phonology that represents a watershed in the history of the language with compared with EP, which follows another pathway (Bisol 1981). Vowel harmony is an assimilatory process that affects pretonic vowels /e, o/ in correspondence with high vowels. Other cases of harmony include pretonic vowel lowering, depending on the height of the vowel in the following syllable, as in *colega* [kɔˈlegɐ] ‘colleague’. Pretonic vowel raising may also occur with no apparent phonetic motivation, as in *governo* [guˈverno] and *boneca* [buˈnekɐ], among others.

Silva Neto (1970) and Camara Jr. (1975) are among the first studies to mention vowel harmony in BP, while Bisol (1981) was the first to provide a thorough description and analysis, using data from Southern BP, in which the frequency of use of pretonic vowel harmony appears stable at 35%, despite recent increases in Rio Grande do Sul (Schwindt 2002). Bisol (1981) found that vowel harmony is conditioned mainly by linguistic factors. The raising of /e/ toward [i] (e.g., *p[i]pino* ‘cucumber’, *p[i]ludo* ‘hairy’) occurs more often than that of /o/ toward [u] (e.g., *c[u]stume* ‘custom’, *c[u]zinha* ‘kitchen’), regardless of whether the high vowel in the following syllable is coronal or dorsal. Bisol (1981) explains this difference in frequency based on the fact

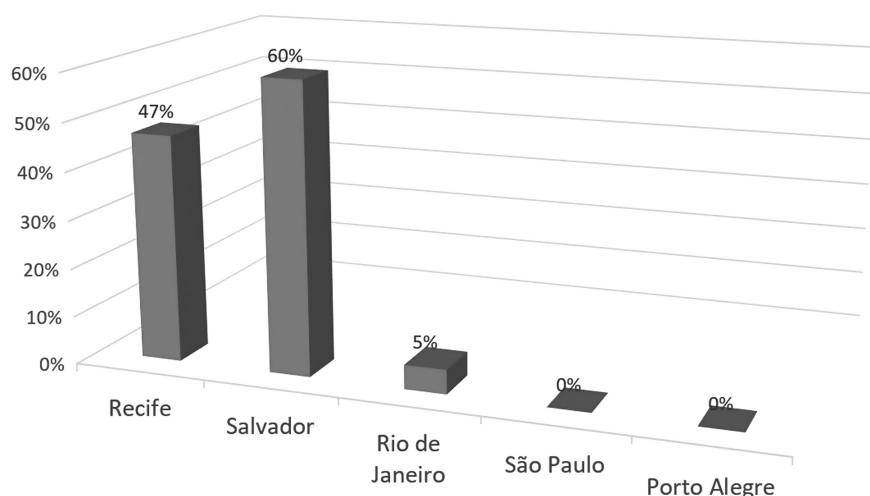


Figure 10.1 Percentages of occurrence of open mid (pretonic) vowels in the speech of cities surveyed in the NURC project (based on Callou & Leite 2002: 40).

that /i/ is higher than /u/. The homorganicity of features between target and trigger vowels plays a role, and greater assimilation is observed between coronal vowels [e] and [i]. Schwindt (2002) confirms the findings of Bisol's (1981) study with data from Rio Grande do Sul as part of the VARSUL project.

Bisol (2010) examined pretonic vowel raising in words without high vowels. The author's hypothesis was that raising in this case was phonetic in nature and, thus, differed the raising caused by vowel harmony. Bisol (2010) argues that both are different formal processes, as "vowel reduction without a specific phonetic conditioning factor is clearly a diffusing process which spreads through the lexicon through a group of words sharing a common base" (Bisol 2010: 78).⁷

Pretonic vowels /e, o/ are also realized as either [e, o] or [ɛ, ɔ], respectively, depending on the dialectal area. Indeed, this variable pronunciation is considered a delimiting characteristic of BP dialects, that is, "a dividing line between northern speech, which opts for open mid vowels, and southern speech, which opts for closed mid vowels" (Leite & Callou 2002: 38). Figure 10.1 shows the percentages of occurrence of open mid (pretonic) vowels in the speech of the five cities surveyed in the NURC project.

As illustrated in Figure 10.1, Salvador and Recife lead the production of pretonic /e, o/ as [ɛ, ɔ], while their occurrence in Rio de Janeiro is marginal and non-existent in São Paulo and Porto Alegre. This scenario remains unchanged (cf. Brandão 2015).

4 Concluding remarks

Providing an exhaustive account of sound variation in BP remains a difficult undertaking, despite the many advances in sociolinguistics, dialectology, and phonology in Brazil in the past 50 years. To date, there are no truly comprehensive studies on all variable phenomena in BP phonology, covering all BP dialects. The use of different methodologies and analytical techniques also contributes to a limitation in comprehensive research findings. After decades of research and data collection, the question of 'how do Brazilians speak' is still in need of a comprehensive answer. Indeed, this

very question was posed as the title of Leite and Callou's (2002) volume, in which they argued that BP varieties represent "unity in diversity and diversity in unity."

Linguistic diversity in Brazil is the result of cultural plurality, different patterns of colonization and population distribution, mixing of different ethnicities, and contact among different languages. This complexity, coupled with the size of Brazil's territory, makes it "practically impossible to provide a complete description of the dialectal and social variation that resulted from contact between Portuguese and other languages and from the changes that were brought to here" (Leite & Callou 2002: 13).⁸ However, there are many pathways that should be explored in future research, such as reexamining real-time variable processes with new data, expanding research on Midwestern and Northern BP varieties, comparing results from sociolinguistic and geolinguistic studies, etc.

Notes

- 1 Variation in BP at the suprasegmental level is beyond the scope of this chapter (cf. Chapters 6–9).
- 2 Reference to *r*-sounds vary considerably in the literature (e.g., 'rhotic', 'vibrant', 'non-lateral liquid'). In this chapter, we choose to represent this group of sounds with the archiphoneme /R/ (cf. Camara Jr. 1970).
- 3 Translated from: "Não pode ser considerada um traço popular restrito a algumas áreas, como Ceará e Rio de Janeiro, como se poderia pensar, a julgar por textos clássicos. Comparando a fala culta de São Paulo, Rio de Janeiro, Salvador, Recife e Porto Alegre, é fácil perceber que Porto Alegre se distingue das demais, sendo a realização alveolar/velar praticamente exclusiva dessa capital" (Leite & Callou 2002: 47).
- 4 Translated from: "uma entre as variantes lateral e rótico e outra entre as variantes lateral e vocalização. Por exemplo, em São José do Norte, dos 8 informantes que mais vocalizam, 6 não apresentam nenhuma ocorrência de rótico. Por outro lado, naqueles que rotacizam, encontramos baixas taxas de vocalização" (Collischonn & Quednau 2010: 135).
- 5 De Paula (2010: 70) analyzed data from different corpora, namely, the NURC project, the *Programa de Estudos sobre o Uso da Língua* [Program Language Use Studies] (PEUL), the *Atlas Etnográfico dos Pescadores do Rio de Janeiro* [Ethnographic Atlas of Fishermen in Rio de Janeiro] (APERJ). Data from 18 informants were collected from the NURC surveys, while 78 fishermen were surveyed in the APERJ, and 25 interviews were conducted for the PEUL.
- 6 Ribeiro (2007: 138) uses various methods of data collection, such as questionnaires, guided and spontaneous interviews, etc. Among analyzed contexts are "occasional proparoxytones," which are words with diphthongs at the end of a word (e.g., *falácia* 'fallacy', *câmbio* 'exchange'). This probably explains their high number of occurrence (1,823 tokens).
- 7 Translated from: "a redução vocálica sem condicionador fonético específico dá sinais claros de ser um processo difusionista, cujo canal de expansão é o léxico por onde se estende via grupo de palavras com uma base em comum" (Bisol 2010: 78).
- 8 Translated from: "praticamente impossível traçar um quadro completo de variação dialetal e socialmente diferenciada resultante do contato com outras línguas e, por sua vez, de mudanças ocorridas na língua portuguesa trazida para cá" (Callou & Leite 2002: 13).

Further reading

- Bisol, Leda and Cláudia Brescancini (eds.). 2002. *Fonologia e variação: Recortes do português brasileiro*. Porto Alegre: EDIPUCRS.
- Hora, Dermeval da, Elisa Battisti, and Valéria Monaretto (eds.). 2019. *Mudança fônica do português brasileiro*. São Paulo: Contexto.
- Martins, Marco Antônio and Jussara Abraçado (eds.). 2015. *Mapeamento sociolinguístico do português brasileiro*. São Paulo: Contexto.

References

- Altman, Cristina. 2012. História, estórias e historiografia da linguística brasileira. *Todas as Letras* 14.14–37.
- Amaral, Marisa Porto. 2000. *As proparoxítonas: Teoria e variação*. PhD. dissertation, Pontifical Catholic University of Rio Grande do Sul.
- Amaral, Marisa Porto. 2002. A síncope em proparoxítonas: Uma regra variável. In Leda Bisol and Gisela Collischonn (eds.), *Português do sul do Brasil: Variação fonológica*. Porto Alegre: EDIPUCRS, 99–126.
- Araújo, Maria Francisca Ribeiro de. 1999. *A alternância de [ei] ~ [e] no português falado na cidade de Caxias*. Master's thesis, Campinas State University.
- Bailey, Guy and Jan Tillery. 2004. Some sources of divergent data in sociolinguistics. In Carmen Fought (ed.), *Sociolinguistic Variation: Critical Reflections*. New York: Oxford University Press, 11–30.
- Battisti, Elisa. 2002. A redução dos ditongos nasais átonos. In Leda Bisol and Cláudia Brescancini (eds.), *Fonologia e variação: Recortes do português brasileiro*. Porto Alegre, EDIPUCRS, 183–202.
- Battisti, Elisa and Viviane T. Moras. 2016. A vocalização da consoante lateral em coda silábica em uma variedade de português brasileiro: Análise sociolinguística em tempo real. *Gragoatá*. 21.40.
- Bertani, Sílvia Renata. 1998. Análise fonológica do infinitivo na fala de Porto Alegre. *Revista Língua e Literatura* 1.19–40.
- Bisol, Leda. 1981. *Harmonização vocálica: Uma regra variável*. Ph.D. dissertation, Federal University of Rio de Janeiro.
- Bisol, Leda. 1989. O ditongo na perspectiva da fonologia atual. *D.E.L.T.A.* 5/2.185–224.
- Bisol, Leda. 1994. Ditongos Derivados. *D.E.L.T.A.* 10.123–140.
- Bisol, Leda. 2010. O alargamento da pretônica sem motivação aparente. In Leda Bisol and Gisela Collischonn (eds.), *Português do sul do Brasil: Variação fonológica*. Porto Alegre: EDIPUCRS, 73–92.
- Bisol, Leda and Cláudia Brescancini (eds.). 2002. *Fonologia e variação: Recortes do português brasileiro*. Porto Alegre: EDIPUCRS.
- Brandão, Sílvia Figueiredo. 2015. Variação e mudança no âmbito do vocalismo. In Marco Antônio Martins and Jussara Abraçado (eds.), *Mapeamento sociolinguístico do português brasileiro*. São Paulo: Contexto, 11–38.
- Brandão, Sílvia Figueiredo. 2022. Desafios da constituição de corpora linguísticos. In Raquel Meister Freitag, Silvana Silva de Farias Araújo, and Valter de Carvalho Dias (eds.), *Desafios para a pesquisa em sociolinguística*. São Paulo: Blucher.
- Cabreira, Sílvia H. 1996. *A monotongação dos ditongos orais decrescente em Curitiba, Florianópolis e Porto Alegre*. Master's thesis, Pontifical Catholic University of Rio Grande do Sul.
- Cagliari, Luiz Carlos. *Elementos de fonética do português brasileiro*. São Paulo: Paulistana, 2007.
- Callou, Dinah. 1987. *Variação e distribuição da vibrante na fala culta carioca*. Ph.D. dissertation, Federal University of Rio de Janeiro.
- Callou, Dinah. 2008. Revisitando o -R. In Cláudia Roncarati and Sebastião Votre (eds.), *Anthony Julius Naro e a Linguística no Brasil*. Rio de Janeiro: 7 Letras. 135–146.
- Callou, Dinah, Yonne Leite, and João Moraes. 2002. Processo(s) de enfraquecimento consonantal no português do Brasil. In Maria Bernadete Abaurre and Angela C. S. Rodrigues (eds.), *Gramática do português falado VIII: Novos estudos descritivos*. Campinas: Editora da Unicamp, 537–555.
- Callou, Dinah, João Moraes, and Yonne Leite. 1996. Variação e diferenciação dialetal: A pronúncia do /R/ no português do Brasil. In Ingedore G. Villaça Koch (ed.), *Gramática do Português Falado, VI: Desenvolvimentos*. Campinas: Editora da Unicamp, 465–493.
- Callou, Dinah and Carolina Serra. 2012. Variação do rótico e estrutura prosódica. *Revista do GEL* 14.41–57.
- Callou, Dinah, Carolina Serra, and Cláudia Cunha. 2015. Mudança em curso no português brasileiro: O apagamento do R no dialeto nordestino. *Revista da ABRALIN* 14.195–219.
- Camara Jr. Joaquim Mattoso. 1971a. *História da linguística*. Petrópolis: Vozes.
- Camara Jr. Joaquim Mattoso. 1971b. *Problemas de linguística descritiva*. 12th edn. Petrópolis: Vozes.
- Camara Jr. Joaquim Mattoso. 1975. *Estrutura da Língua Portuguesa*. Petrópolis: Vozes.
- Camara Jr. Joaquim Mattoso. 1977. *Para o estudo da fonêmica portuguesa*. Rio de Janeiro: Padrão.
- Castilho, Ataliba Teixeira de. 2007. Prefácio. In Anthony J. Naro and Maria Marta Pereira Sherre (eds.), *Origens do português brasileiro*. São Paulo: Parábola, 11–15.
- Chaves, Raquel. 2017. *A redução/desnasalização de ditongos nasais átonos finais e a marcação explícita de CVP6: Um estudo de correlação*. Ph.D. dissertation, Federal University of Santa Catarina.

- Collischonn, Gisela and Laura R. Quednau. 2010. As laterais variáveis da Região Sul. In Leda Bisol and Gisela Collischonn (eds.), *Português do sul do Brasil: Variação fonológica*. Porto Alegre: EDIPUCRS, 128–146.
- Cristofolini, Carla. 2011. Estudo da monotongação de [ow] no falar florianopolitano: Perspectiva acústica e sociolinguística. *Revista da ABRALIN* 10.205–229.
- Cunha, Celso. 1986. Conservação e inovação no português do Brasil. *O Eixo e a Roda: Revista de Literatura Brasileira* 5.199–230.
- De Paula, Alessandra. 2010. *Vogais médias postônicas na fala do estado do Rio de Janeiro*. Master's thesis, Federal University of Rio de Janeiro.
- Espiga, José W. R. 2001. *O português dos campos neutrais: Um estudo sociolinguístico da lateral pós-vocálica nos dialetos fronteiriços do Chuí e Santa Vitória do Palmar*. Ph.D. dissertation, Pontifical Catholic University of Rio Grande do Sul.
- Gregis, Hilaine. 2001. *O apagamento da vibrante pós-vocálica em Porto Alegre*. Master's thesis, Federal University of Rio Grande do Sul.
- Guiotti, Luciana Prudente. 2002. *O estudo da variante retroflexa na comunidade de São José do Rio Preto*. Master's thesis, São Paulo State University, São José do Rio Preto.
- Guy, Gregory. 1981. *Linguistic Variation in Brazilian Portuguese: Aspects of the Phonology, Syntax and Language History*. Ph.D. dissertation, University of Pennsylvania.
- Hora, Dermeval da and Valéria Neto de Oliveira Monaretto. 2003. Enfraquecimento e apagamento dos róticos. In Dermeval da Hora and Gisella Collischonn (eds.), *Teoria linguística: Fonologia e outros temas*. João Pessoa: Editora Universitária, 114–143.
- Hora, Dermeval da, Elisa Battisti, and Valéria Neto de Oliveira Monaretto. 2019. *História do português brasileiro. Volume 3: Mudança fônica do português brasileiro*. São Paulo: Contexto.
- Hora, Dermeval da and W. Leo Wetzels. 2010. Róticos: Uma “po[h, r, ʁ]ta” entre paraibanos e paulistanos. *Linguística: Revista da Associação de Linguística e Filologia da América Latina* 24.51–76.
- Leal, Eneida Goes. 2016. *Os róticos no Varsul: Uma generalização de dados*. Porto Alegre: ReVEL 13.292–319.
- Lee, Seug Hwa (ed.). 2012. *Vogais além de Belo Horizonte*. Belo Horizonte: FALE/UFGM.
- Leite, Cândida Mara Britto. 2010. *O /R/ em posição de coda silábica no falar campineiro*. Ph.D. dissertation, Campinas State University.
- Leite, Yonne and Dinah Callou. 2002. *Como falam os brasileiros*. Rio de Janeiro: Jorge Zahar Editor.
- Milesky, Ivanete. 2013. *A elevação das vogais médias átonas finais no português falado por descendentes de imigrantes poloneses em Vista Alegre do Prata, RS*. Ph.D. dissertation, Pontifical Catholic University of Rio Grande do Sul.
- Monaretto, Valéria Neto de Oliveira. 2000. O apagamento da vibrante pós-vocálica nas capitais da fala do sul do Brasil. *Letras de Hoje* 35.275–284.
- Monaretto, Valéria Neto de Oliveira. 2002. A vibrante pós-vocálica em Porto Alegre. In Leda Bisol and Cláudia Brescancini (eds.), *Fonologia e variação: Recortes do português brasileiro*. Porto Alegre: EDIPUCRS, 253–268.
- Monaretto, Valéria Neto de Oliveira. 2010. Descrição da vibrante no português do sul do Brasil. In Leda Bisol and Gisela Collischonn (eds.), *Português do sul do Brasil: Variação fonológica*. Porto Alegre: EDIPUCRS, 141–151.
- Monaretto, Valéria Neto de Oliveira. 2013. Valor fonético das vogais médias postônicas em jornais oitocentistas gaúchos. *Diadorim* 14.77–98.
- Monaretto, Valéria Neto de Oliveira. 2019. Sequências mediais de obstruintes. In Dermeval da Hora, Elisa Battisti, and Valéria Neto de Oliveira Monaretto (eds.), *História do português brasileiro. Volume 3: Mudança fônica do português brasileiro*. São Paulo: Contexto, 138–157.
- Naro, Anthony J. 1973. *Estudos diacrônicos*. Petrópolis: Vozes.
- Oliveira, Marco Antônio. 1983. *Variation and Change in Brazilian Portuguese: The Case of the Liquids*. Ph.D. dissertation, University of Pennsylvania.
- Oushiro, Livia and Ronald Beline Mendes. 2013. A pronúncia do (–r) em coda silábica no português paulistano. *Revista do GEL* 8.66–95.
- Paiva, Maria da Conceição Auxiliadora de. 1996. Supressão de semivogais nos ditongos decrescentes. In Giselle Machline de Oliveira e Silva and Maria Marta Pereira Scherre (eds.), *Padrões Sociolinguísticos: Análise de fenômenos variáveis na cidade do Rio de Janeiro*. Rio de Janeiro: Tempo Brasileiro.

- Quednau, Laura Rosane. 1993. *A lateral pós-vocálica no português gaúcho: Análise variacionista e representação não-linear*. Master's thesis, Federal University of Rio Grande do Sul.
- Rennicke, Iris. 2011. The retroflex r of Brazilian Portuguese: Theories of origin and a case study of language attitudes in Minas Gerais. *Revista de Estudos Linguísticos da Universidade do Porto* 6.149–170.
- Révah, I. S. 1958. L'évolution de la prononciation au Portugal et au Brésil du XVI^e siècle à nos jours. *Anais do Primeiro Congresso de Língua Falada no Teatro, Salvador, 1956*. Rio de Janeiro: Ministério da Educação e Cultura, 387–399.
- Ribeiro, Darinka Fortunato Suckow. 2007. *Alçamento de vogais postônicas não finais no português de Belo Horizonte: Uma abordagem difusionista*. Master's thesis, Federal University of Minas Gerais.
- Rodrigues, Sandra Arouca. 2001. *O -s pós-vocálico na fala da região norte-noroeste do estado do Rio de Janeiro*. Master's thesis, Federal University of Rio de Janeiro.
- Schwindt, Luiz Carlos. 2002. A regra variável de harmonização sonora no RS. In Leda Bisol and Cláudia Brescancini (eds.), *Fonologia e variação: Recortes do português brasileiro*. Porto Alegre: EDIPUCRS, 165–182.
- Schwindt, Luiz Carlos and Raquel Gomes Chaves. 2019. Convergência de processos no apagamento de /R/ em português e espanhol. *Linguística* 35.129–147.
- Schwindt, Luiz Carlos e Tais Bopp da Silva. 2010. Panorama da redução da nasalidade em ditongos átonos finais no português do sul do Brasil. In Leda Bisol and Gisela Collischonn (eds.), *Português do sul do Brasil: Variação fonológica*. Porto Alegre: EDIPUCRS, 15–30.
- Secco, Glacy C. 1977. *O // implosivo na linguagem pontagrossense*. Master's thesis, Federal University of Rio de Janeiro.
- Silva Neto, Serafim. 1970. *História da língua portuguesa*. 2nd edn. Petrópolis: Vozes.
- Silveira, Giselle. 2010. *O apagamento da vibrante na fala do sul do Brasil: Sob ótica da palavra*. Master's thesis, Federal University of Rio Grande do Sul.
- Skeete, Nadir Arruda. 1996. *Análise variável da vibrante na fala de João Pessoa*. Master's thesis, Federal University of Paraíba.
- Tasca, Maria. 1999. *A lateral em coda silábica no Sul do Brasil*. Ph.D. dissertation, Pontifical Catholic University of Rio Grande do Sul.
- Tasca, Maria. 2002. Variação e mudança do segmento lateral na coda silábica. In Leda Bisol and Cláudia Brescancini (eds.), *Fonologia e variação: Recortes do português brasileiro*. Porto Alegre, EDIPUCRS, 269–302.
- Toledo, Eduardo Elisalde. 2011. *A monotongação do ditongo decrescente [ej] na amostra de recontato de Porto Alegre*. Master's thesis, Federal University of Rio Grande do Sul.
- Uchôa, Carlos Eduardo Falcão. 1975. Os estudos e a carreira de Joaquim Mattoso Camara Jr. In Carlos Eduardo Falcão Uchôa (ed.), *Dispersos de J. Mattoso Camara Jr.* Rio de Janeiro: Fundação Getúlio Vargas.
- Vieira, Maria José Blakovski. 2010. As vogais médias átonas nas três capitais do sul do país. In Leda Bisol and Gisela Collischonn (eds.), *Português do sul do Brasil: Variação fonológica*. Porto Alegre: EDIPUCRS, 50–72.
- Votre, Sebastião Josué. 1978. *Aspectos da variação fonológica na fala dos analfabetos do Rio de Janeiro*. Ph.D. dissertation, Pontifical Catholic University of Rio de Janeiro.
- Weinreich, Uriel, William Labov, and Marvin Herzog. 1968 [2006]. *Fundamentos empíricos para uma teoria da mudança linguística*. São Paulo: Parábola.
- Williams, Edwin B. 1975. *Do latim ao português – Fonologia e morfologia históricas da língua portuguesa*. 3rd edn. Rio de Janeiro: Tempo Brasileiro.